A REPORT ON PROJECT WORK TITLED

TOURISM GUIDE

SUBMITTED BY PROJECT

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IN PARTIAL FULFILMENT OF REQUIREMENTS FOR THE AWARD OF DIPLOMA IN

COMPUTER ENGINEERING

UNDER THE ESTEEMED GUIDENCE OF

Smt.N. Pavani

DEPARTMENT OF COMPUTER ENGINEERING

ANDHRA POLYTECHNIC KAKINADA - 533002

2022-2025

DEPARTMENT OF COMPUTER ENGINEERING

CERTIFICATE

This is to certify that this project work entitled "**Tourism Guide**" is the Bonafide record work done by **Mr./Ms.Akasapu.Manjusha** bearing the **Pin N0 22010-CM-004** of the final year, along with batch mates submitted on the partial fulfilment of the requirement for the award of **DIPLOMA IN COMPUTER ENGINEERING** during the academic year 2022-2025.

Signature of project guide
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Signature of the External Examiner

ACKNOWLEDGEMENT

We have been tried to our level best to complete this project with group discussion among our batch- mates and taking valuable suggestions from our classmates and made it to work properly and effectively.

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We also thankful to our Head of Computer Engineering and our project coordinator **Sri. G.Partha Sarthi (M.Tech),** for his cooperation and unflinching Devotion, which motivated us to complete this project.

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TABLE OF CONTENT

ABSTRACT

Travel Desi is an online India tour guide platform that helps travelers explore and book tour guides, hotels, and restaurants easily. The platform provides a structured way to navigate India's famous tourist destinations, dividing them into four regions: **East, West, North, and South**. Users can explore the top 10 famous places in each region, and upon selecting a location, they will get detailed information, including famous spots, nearby guides, hotels, and restaurants. The platform also allows direct booking of guides, making travel planning more seamless and efficient. Additionally, the website integrates Google Translator, enabling travelers from different linguistic backgrounds to access content in their preferred language.

EXISTING SYSTEM

In the current travel industry, travelers rely on multiple platforms to plan trips, including searching for guides, booking hotels, and finding restaurants separately. This process is time-consuming and lacks a unified system for connecting tourists with trusted guides. Moreover, there is limited integration of local guides who have deep knowledge of the destinations. Most platforms provide only general information without direct booking options for guides, leading to a lack of personalized travel experiences. Language barriers further limit the ease of exploring travel-related content for international and regional travelers.

PROPOSED SYSTEM

Travel Desi aims to overcome the limitations of existing systems by offering a one-stop platform for travel planning. The proposed system includes various key features. The home page provides options for guide registration, hotel registration, and restaurant registration, along with a guide login to manage bookings. Users can directly book guides and explore a display of bucket-list travel destinations. The platform is structured into a regional travel module, where India is divided into four regions: East, West, North, and South. Each region showcases the top 10 most famous tourist places, and clicking on a place navigates to its dedicated page. On the place details page, users can view six famous attractions within the selected location, access integrated Google Maps for navigation, and find nearby available guides, restaurants, and hotels with contact details. To enhance accessibility, the website includes Google Translator integration, allowing users to translate content into their preferred language. This feature helps international and regional travelers overcome language barriers, ensuring an easy and seamless travel experience.

SOFTWARE REQUIREMENTS

SOFTWARE SPECIFICATION:

OS : MS WINDOWS 11

Front End : HTML, CSS, React Js, Boot Strap

Back End : Node js, Express js

Database : MongoDB

Platform : Visual Studio Code

HARDWARE SPECIFICATION :

Processor : Intel Core i5 13th Gen with iRisxe Graphics 2.10 GHz

RAM : 16.0 GB RAM

Hard Disk : 512 GB

Keyboard : Standard keyboard

Mouse : Standard mouse

Monitor : Standard color

WEB BROWSER:

CHROME, INTERNET EXPLORER (OR) ANY BROWSER

INTRODUCTION

Travel Desi is an innovative online tour guide platform designed to enhance travel experiences across India. The platform serves as a one-stop solution for tourists, allowing them to explore destinations, book local guides, and find accommodations and restaurants all in one place. Unlike conventional travel websites that provide only general information, Travel Desi offers direct access to travel services, making trip planning more efficient and hassle-free.

The website categorizes India into four regions—East, West, North, and South—allowing travelers to explore famous locations within each region. Each region showcases the top 10 must-visit tourist destinations, and clicking on any place navigates users to a dedicated page that provides details about the site, including six famous attractions, integrated Google Maps for easy navigation, and contact information for nearby guides, hotels, and restaurants.

One of the key features of Travel Desi is its direct guide booking system, which allows tourists to connect with verified guides for a more personalized travel experience. This eliminates the need for third-party intermediaries, ensuring authenticity and reliability. Additionally, the platform integrates Google Translator, enabling users to access content in their preferred language, breaking language barriers for both domestic and international tourists.

By combining essential travel services with advanced features such as real-time location tracking and multilingual support, Travel Desi aims to revolutionize travel planning in India. Whether a traveler is exploring a well-known destination or looking for hidden gems, the platform provides a seamless and user-friendly experience, making travel more accessible, informative, and enjoyable.

Technologies Used in Travel Desi

Front-End Technologies

HTML (HyperText Markup Language)

is used to create the structure of web pages, ensuring content is organized in a meaningful way. It defines essential elements like headings, paragraphs, images, links, buttons, and input forms to make the website functional.

CSS (Cascading Style Sheets)

adds styling and aesthetics to the website by defining colors, fonts, layouts, and animations. It ensures responsiveness, making the application adaptable to different screen sizes and devices, improving user experience.

React.js

is a JavaScript library that enhances the front-end with a dynamic and component-based architecture. It enables fast rendering using the virtual DOM, allowing smooth page transitions, interactive UI elements, and an optimized user experience.

Bootstrap

is a widely used CSS framework that provides pre-built UI components such as modals, buttons, forms, and navigation bars. It helps in designing a mobile-friendly and visually appealing interface without the need for extensive custom CSS styling.

Back-End Technologies

Node.js

Node.js is a **JavaScript runtime environment** that enables server-side scripting. It follows an **event-driven, non-blocking architecture**, making it ideal for handling real-time applications. Node.js efficiently processes API requests and manages data flow between the front end and the database.

Express.js

Express.js is a lightweight Node.js framework that simplifies back-end development. It provides **middleware integration**, **routing capabilities**, and **RESTful API support**, allowing seamless communication between the front end and database.

Python

Python is widely used in **backend development, automation, and data processing**. With frameworks like **Django and Flask**, it enables secure, scalable, and efficient server-side applications. Python is also leveraged for **data analytics and scripting tasks** in web applications.

Database Management

MongoDB

MongoDB is a **NoSQL database** that stores data in a flexible, JSON-like format. It handles **structured and unstructured data** efficiently, making it ideal for real-time applications. Features like **sharding and replication** improve scalability and reliability

SYSTEM ANALYSIS FOR THE TOURISM GUIDE

The Tourism Guide System is designed to enhance the travel experience by providing tourists with accurate and reliable information about destinations, accommodations, restaurants, and local guides. The system ensures seamless trip planning by integrating essential travel services into a single platform.

The main objectives of the system are to provide an easy-to-navigate platform for tourists to explore destinations, enable direct service provider registration for guides, hotels, and restaurants, ensure multilingual accessibility using a translation module, and offer real-time search and navigation capabilities to enhance travel experiences.

Traditional tourism websites provide general information but lack personalized services, real-time guide booking, and direct communication with service providers. Additionally, language barriers often make it difficult for international tourists to access relevant details. The Tourism Guide System addresses these inefficiencies by offering an interactive and service-oriented approach.

The system includes several essential functionalities. User registration and authentication allow tourists, hotels, restaurants, and guides to register and log in securely. Secure authentication ensures that only verified users can access services. Destination management categorizes destinations into four regions: East, West, North, and South. Each region features ten must-visit places with detailed descriptions, travel tips, and local guides. Users can explore six key attractions within each destination, including images and highlights.

Service provider registration enables hotels, restaurants, and tour guides to register and list their services for better accessibility. Tourists can view service ratings and contact service providers directly. The translation module supports multilingual interactions through both text-based and speech-based translations. It integrates the Google Translator API to translate travel-related content for international tourists.

The search and navigation feature allows users to search destinations, guides, accommodations, and restaurants using keywords. The integrated Google Maps API provides accurate location tracking and navigation assistance. It enables real-time route planning and travel distance estimation.

The system offers several benefits, including an enhanced user experience by providing personalized travel recommendations based on user preferences. It eliminates third-party intermediaries, enabling direct tourist-to-guide interactions. It simplifies trip scheduling with real-time service availability. The multilingual accessibility feature removes language barriers, making the platform user-friendly for global travelers.

By addressing existing inefficiencies and integrating advanced technologies, the Tourism Guide System aims to revolutionize travel planning and navigation in India.

Functional Requirements of Travel Desi

User Registration & Authentication: Secure registration and login for tourists, guides, hotels, and restaurants, allowing profile management.

Destination Management: India is categorized into four regions—East, West, North, and South—each featuring ten tourist destinations with details, attractions, nearby services, and Google Maps integration.

Guide Booking System: Tourists can directly book verified local guides, ensuring authenticity and seamless booking without third-party involvement.

Service Provider Registration: Hotels, restaurants, and travel service providers can register, displaying business details, contact information, and customer ratings.

Translation Module: Google Translator API enables multilingual support, making the platform accessible to international travelers.

Search & Navigation: A powerful search feature and Google Maps integration help users find destinations, guides, and services efficiently.

Security & Authentication: JWT-based authentication ensures secure user data management and prevents unauthorized access.

Responsive & User-Friendly Interface: Built with React.js and Bootstrap, the platform ensures smooth navigation and mobile-friendly accessibility.

FEASIBILITY STUDY FOR TRAVEL DESI

The feasibility study for Travel Desi evaluates whether the project is practical and achievable within available resources, technology, and budget. It ensures that the system can be developed and used effectively without unnecessary costs or complexities. The study is divided into three main areas: **economic feasibility, technical feasibility, and operational feasibility.**

1. Economic Feasibility

Economic feasibility assesses whether the cost of developing the system is reasonable compared to the expected benefits.

Estimated Costs:

- **Development Costs:** Expenses for designing and coding the system using **HTML, CSS, JavaScript, React.js, Django, PostgreSQL, and Python.**
- **Hosting & Deployment:** Costs related to web hosting, database management, and cloud services.
- **Maintenance Costs:** Future updates, security patches, and performance improvements.

Expected Benefits:

- **Cost Savings:** Reduces the dependency on physical maps, travel agents, and brochures by providing all travel-related services in one platform.
- **Revenue Generation:** Hotels, restaurants, and guides can register their services, generating income through advertisements and premium listings.
- **Tourism Growth:** Easy access to verified guides and real-time travel information enhances user experience, increasing tourist visits and boosting local businesses.

2. Technical Feasibility

Technical feasibility checks whether the system can be built with the available technology, skills, and infrastructure.

Technology Requirements:

- **Frontend:** HTML, CSS, JavaScript, React.js for a dynamic user interface.
- **Backend:** Django (Python) for server-side operations and PostgreSQL for database management.
- **API Integration:** Google Maps API for navigation and Google Translator API for multilingual support.

Skills & Expertise:

- The development team must have expertise in **web development, API integration, and database management.**
- Advanced features like **AI-based recommendations** can be introduced in future updates.

Scalability:

- The system is designed for future expansion, allowing additional **destinations, languages, and travel services.**

3. Operational Feasibility

Operational feasibility ensures that the system is user-friendly and beneficial for all stakeholders.

User Experience:

- **Tourists:** Can explore destinations, book verified guides, find accommodations, and access travel information easily.
- **Hotels, Restaurants & Guides:** Can register their services, manage their listings, and reach a wider audience.

User Acceptance:

- The system is designed to be intuitive and requires minimal training.
- Its direct service booking model enhances reliability, making it attractive to both tourists and service providers.

Challenges & Solutions:

- **Some users may not be familiar with online booking systems** → Solution: Provide step-by-step guides and video tutorials.
- **Internet connectivity issues in remote areas** → Solution: Offer basic offline access for saved travel details.

SOFTWARE REQUIREMENT & SPECIFICATION FOR TRAVEL DESI

The software requirements for **Travel Desi** include essential tools and frameworks to ensure smooth development, scalability, and maintainability.

Operating System

- **Windows 11** (Development Environment) – Ensures compatibility with development tools and a user-friendly interface.

Front-End Development

- **HTML, CSS, JavaScript** – For structuring and styling the UI.
- **Bootstrap** – Provides responsive design and pre-built components.
- **React.js** – Enables dynamic and interactive UI management.

Back-End Development

- **Django (Python)** – Manages server-side logic and API endpoints efficiently.
- **PostgreSQL** – Chosen for its structured data handling, scalability, and reliability.
- **Google Maps & Translator API** – For navigation and multilingual support.

Development Environment & Tools

- **Visual Studio Code** – A powerful code editor supporting multiple programming languages and extensions.

This setup ensures **efficient travel management**, seamless user experience, and easy scalability.

DESIGN FOR TRAVEL DESI

The system design defines the architecture, components, and modules to ensure efficient functionality and user satisfaction. It includes **Architectural, Logical, and Physical** design aspects.

1. Architectural Design

The system follows a **multi-layered architecture** for scalability and modularity:

User Interface (UI): Allows tourists to explore destinations and use translation features.

Backend API: Manages business logic and data exchange.

Database: Stores user profiles, destinations, guides, hotels, and restaurants.

2. Logical Design

Defines data flow, entity relationships, and module structure:

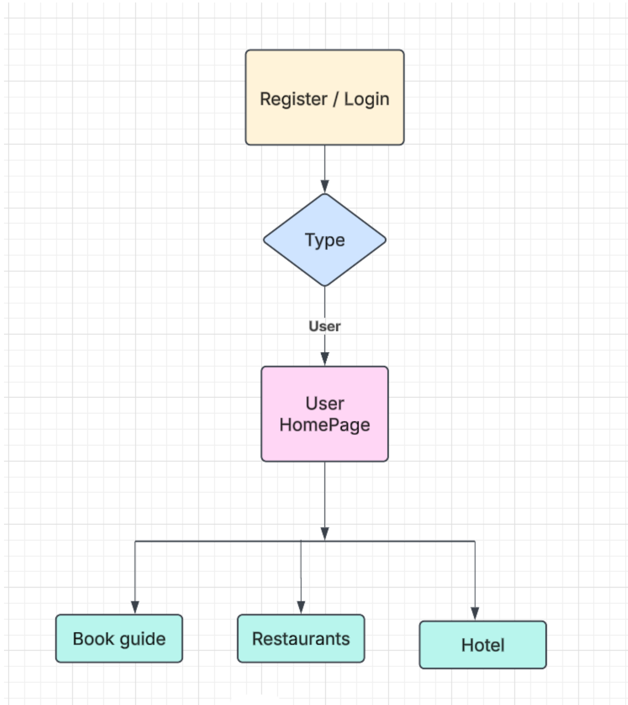
- **Entity-Relationship (ER) Diagrams:** Show relationships between users, places, and services.
- **Data Flow Diagrams (DFD):** Map user actions like searching, viewing details, and booking guides.
- **Modules:**
  - **Tourist Places Module** – Displays destinations and details.
  - **Service Provider Module** – Allows hotels, restaurants, and guides to register.
  - **Guide Module** – Lists available guides with contact details.
  - **Translation Module** – Supports multilingual text and speech translation.
  - **User Authentication Module** – Manages user registration and login.

3. Physical Design

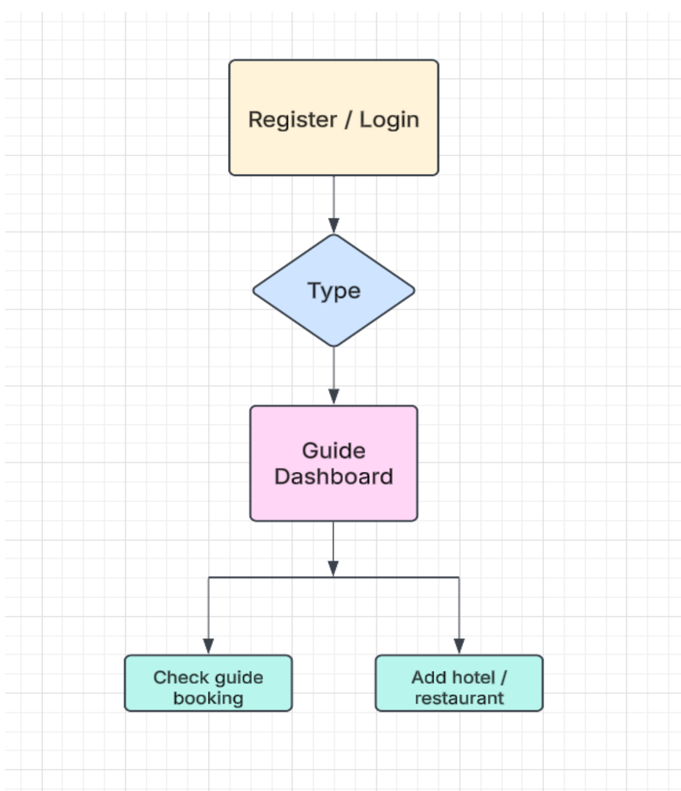
Defines data handling and processing:

- **Input:** Users search, register, and submit reviews.
- **Output:** Displays destinations, hotel details, guides, and translations.
- **Storage:** Securely stores profiles, businesses, and locations.
- **Processing:** Handles user requests, data flow, and AI-driven translations.
- **Security & Backup:** Implements authentication, encryption, and regular backups.

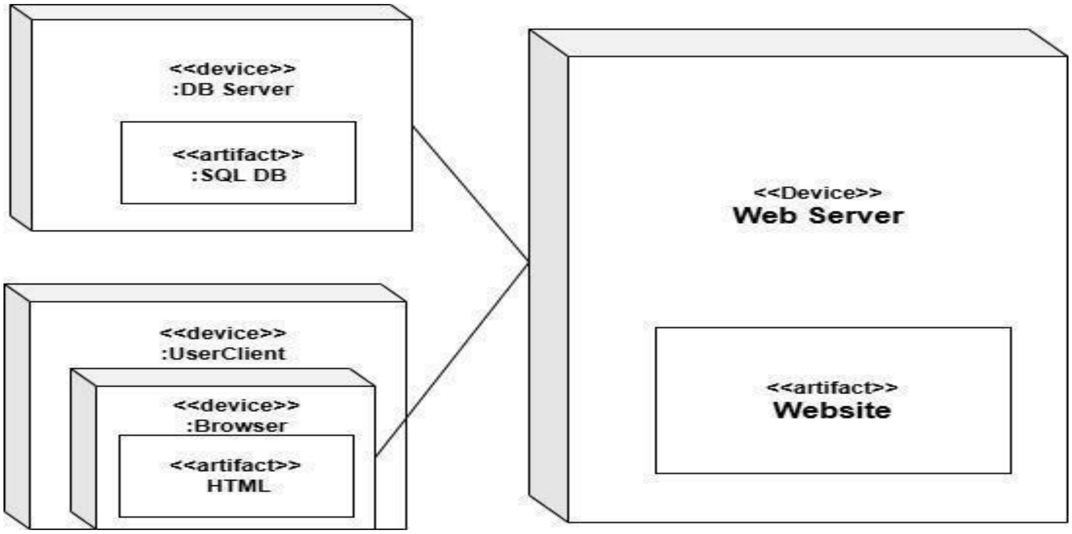
User Module



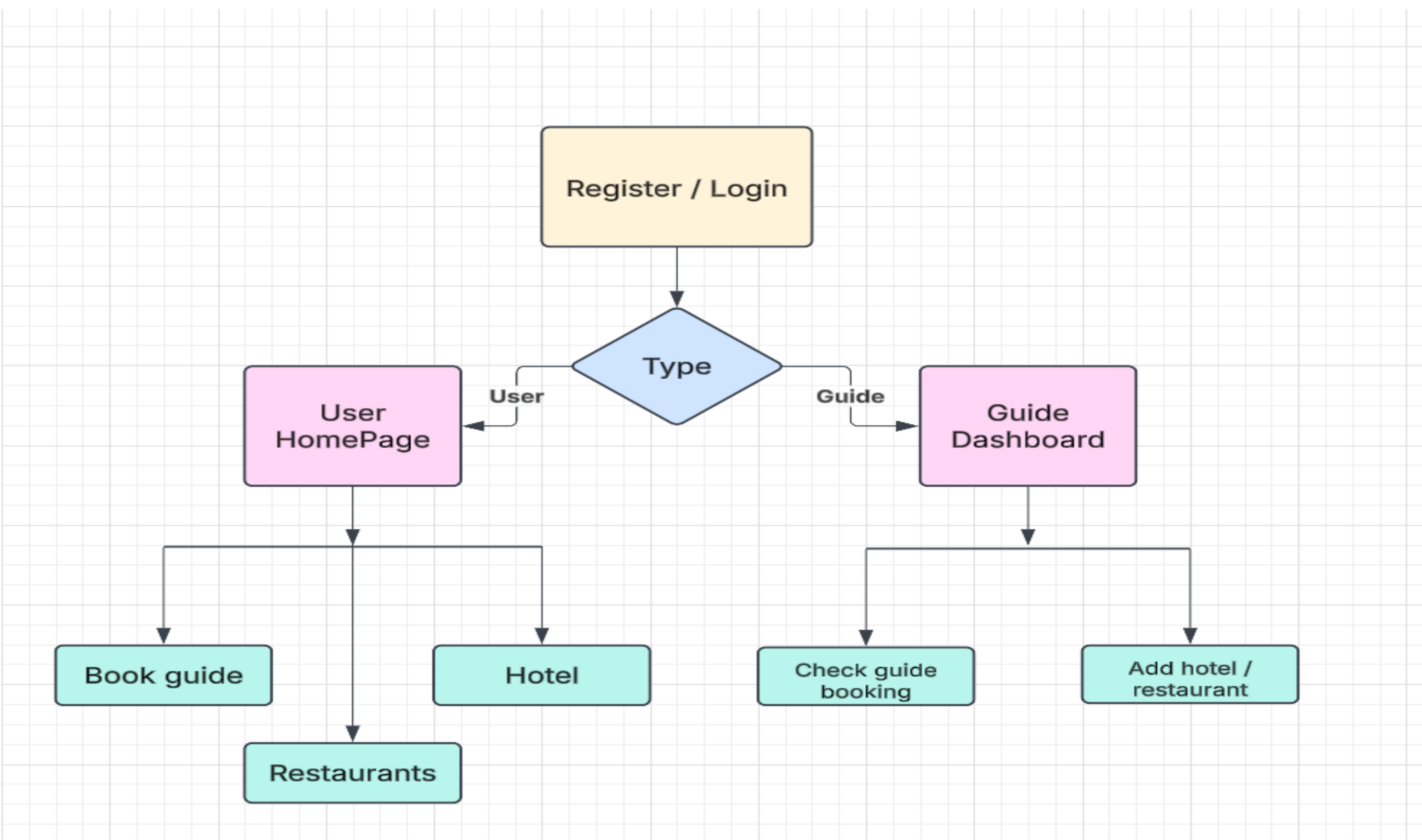
Guide Module



Deployment Diagram



Architectural Design



Windows

Step-by-Step Guide for Installing React.js

Create Project Folder

- Open a terminal or command prompt.
- Navigate to the directory where you want to create your project.

Create a new folder for your React project:

--->cd project

Initialize a New React Project

Use the **npx** command to create a new React application. This automatically sets up a .new React project with all required dependencies

--->npx create-react-app Frontend

Navigate to the Project Directory

Move into the newly created React app folder

--->cd Frontend

Install Additional Dependencies (Optional)

Install any additional packages if your project requires them, like **React Router**, **Axios** (for API calls)
--->npm install react-router-dom axios

Start the Development Server

To check if the installation was successful and to view the default React application in a browser

--->npm start

This command will start the development server and automatically open your application at

http://localhost:3000 in your default web browser.

To Run React.js

```
Microsoft Windows [Version 10.0.22631.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\manjusha>cd desktop

C:\Users\manjusha\Desktop>cd project

C:\Users\manjusha\Desktop\project>cd frontrend

C:\Users\manjusha\Desktop\project\frontrend>npm start

> frontrend@0.1.0 start
> react-scripts start

(node:7480) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is deprecated. Ple
ase use the 'setupMiddlewares' option.
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:7480) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is deprecated. P
lease use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!

You can now view frontrend in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://10.1.40.213:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```

Step-by-Step Guide for Installing and Running MongoDB with Node.js

1. Install Node.js

●Download and install Node.js from Node.js Official Site.
● Verify installation:
node -v
npm -v

2. Install MongoDB
●Download and install MongoDB from MongoDB Official Site.
● Verify installation:
mongod --version
3. Start MongoDB Locally
● Open a terminal and start MongoDB:
mongod
● Open another terminal and start the MongoDB shell:
mongo
● Create a new database:
use tourismDB
4. Install Project Dependencies
● Navigate to your project folder:
cd project
● Install required Node.js packages:
npm install express mongoose cors dotenv

5. Connect to MongoDB in Node.js

● In your project, create a file db.js and add:

```
const mongoose = require('mongoose');
mongoose.connect('mongodb://localhost:27017/tourismDB', {
   useNewUrlParser: true,
   useUnifiedTopology: true,
})
.then(() => console.log("MongoDB Connected"))
.catch(err => console.log(err));
```

6. Run the Backend Server

● Start the server using:

```
node server.js
npm start
```

7. Run MongoDB with MongoDB Atlas (Optional - Cloud Database)

● Replace mongodb://localhost:27017/tourismDB in db.js with the provided MongoDB Atlas URL.

Now your MongoDB and Node.js backend are set up and running successfully!

Coding

In software development,**coding** transforms requirements into functional software, covering activities like designing, writing, testing, and maintaining source code. It includes

1. **Channel & Line Coding:** Improves data transmission accuracy.
2. **Programming:** Core of app development, supported by resources like W3Schools, JavaTpoint, ChatGPT, and YouTube.
3. **Data Coding:** Categorizes quantitative/qualitative data for analysis.
4. **Legal & Therapeutic Coding**: Organizes legal documents and supports addiction treatments.
5. **DNA Coding**: Converts DNA to proteins.
6. **Emergency Codes**: Alerts, e.g., "Code Blue" for cardiac arrest Coding is crucial across domains, enabling efficient data use, categorization, and problem-solving.

Summary About Coding Part

LOGIN PAGE

- Guide
- User

Guide Booking

- User

Add Hotel and Restaurants

Users

Check Bookings

- Guide

TESTING

Testing is a crucial phase in the development of a Pharmacy Management System (PMS) to ensure that the software functions as intended, is free of bugs, and meets user requirements. The testing process typically involves several types of testing, each serving a specific purpose:

Unit Testing:

Objective: Verify the correctness of individual units or components of the system.
Focus: Test individual functions, methods, or procedures.
Example: Test a function that adds a new medicine to the inventory to ensure it properly updates the database.

Integration Testing:

Objective: Ensure that different components/modules of the system work seamlessly together.
Focus: Test interactions between integrated modules.
Example: Test the flow of data between the Inventory Management and Order Processing modules.

System Testing:

Objective: Validate the entire system against specified requirements. **Focus:** Test end- to-end functionalities, user interfaces, and system behavior. **Example:** Test the complete process of placing an order, including customer authentication, medicine availability check, and transaction completion.

Acceptance Testing:

Objective: Ensure that the system meets user expectations and requirements.
Focus: Validate if the system satisfies business use cases.
Example: Have end-users (pharmacists and customers) perform common tasks to confirm usability and functionality alignment.

Performance Testing:

Objective: Assess the system's responsiveness and stability under various conditions.
Focus: Test system scalability, response times, and resource usage.
Example: Simulate a high number of simultaneous transactions to evaluate how the system handles increased load.

Security Testing:

Objective: Identify and address potential vulnerabilities in the system.
Focus: Test for data integrity, user authentication, and protection against common security threats. **Example:** Verify that user authentication mechanisms are robust and that sensitive customer data is appropriately encrypted.

Usability Testing:

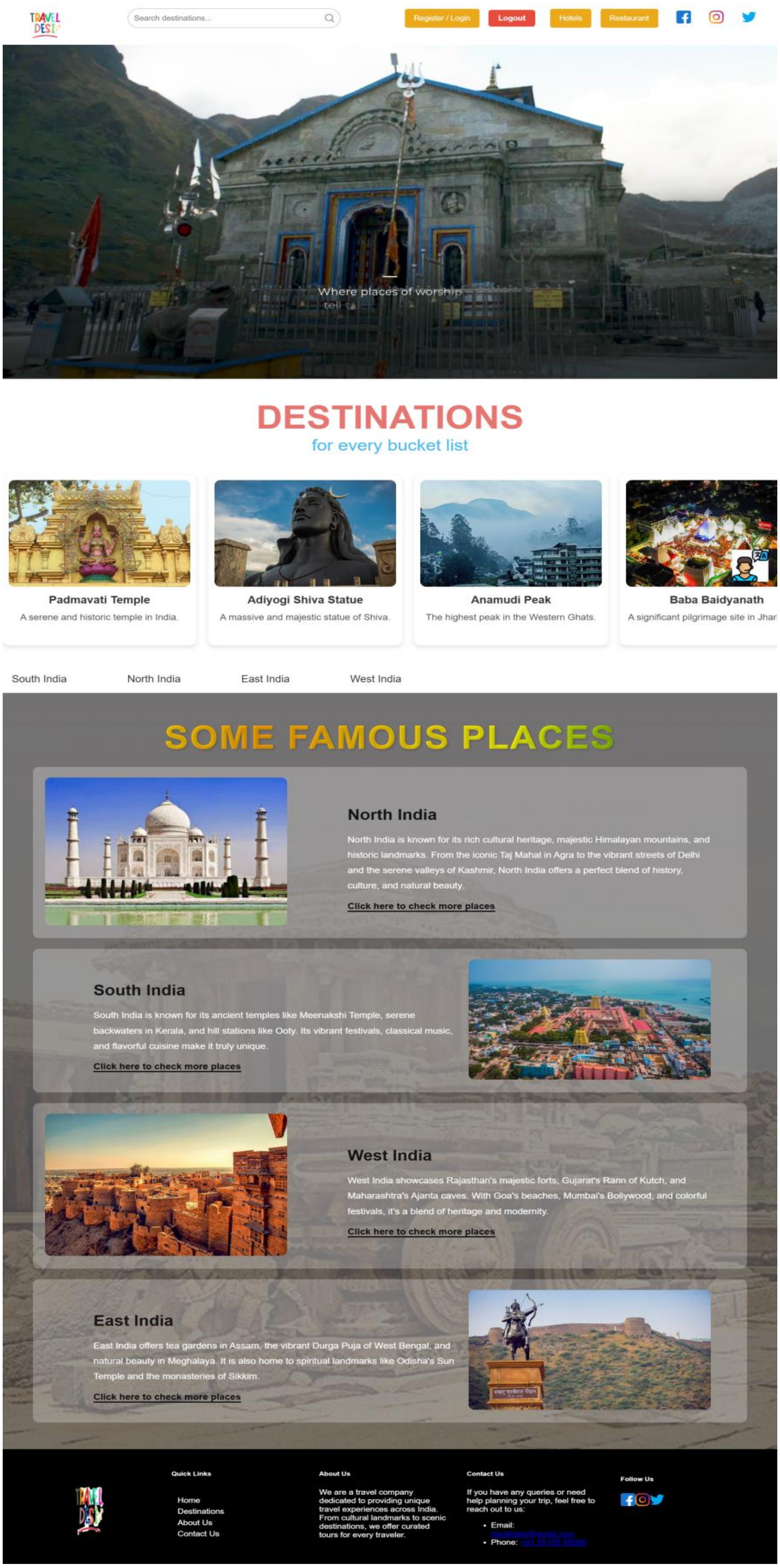
Objective: Evaluate the user interface and overall user experience.
Focus: Assess system navigation, user-friendliness, and accessibility.
Example: Have representative users perform common tasks and provide feedback on the interface design.

Regression Testing:

Objective: Ensure that new changes do not negatively impact existing functionalities.
Focus: Re-test previously validated functionalities after updates or modifications. **Example:** After fixing a bug related to inventory management, ensure that other functionalities are still working as expected.

A comprehensive testing strategy that includes these different types of testing helps ensure the reliability, functionality, and security of the Pharmacy Management System before it is deployed for actual use

HOME PAGE



Search destinations...
Register / Login
Logout
Hotels
Restaurant
Where places of worship
DESTINATIONS
for every bucket list
Padmavati Temple
A serene and historic temple in India.
Adiyogi Shiva Statue
A massive and majestic statue of Shiva.
Anamudi Peak
The highest peak in the Western Ghats.
Baba Baidyanath
A significant pilgrimage site in Jhar
South India
North India
East India
West India
SOME FAMOUS PLACES
North India
North India is known for its rich cultural heritage, majestic Himalayan mountains, and historic landmarks. From the iconic Taj Mahal in Agra to the vibrant streets of Delhi and the serene valleys of Kashmir, North India offers a perfect blend of history, culture, and natural beauty.
Click here to check more places
South India
South India is known for its ancient temples like Meenakshi Temple, serene backwaters in Kerala, and hill stations like Ooty. Its vibrant festivals, classical music, and flavorful cuisine make it truly unique.
Click here to check more places
West India
West India showcases Rajasthan's majestic forts, Gujarat's Rann of Kutch, and Maharashtra's Ajanta caves. With Goa's beaches, Mumbai's Bollywood, and colorful festivals, it's a blend of heritage and modernity.
Click here to check more places
East India
East India offers tea gardens in Assam, the vibrant Durga Puja of West Bengal, and natural beauty in Meghalaya. It is also home to spiritual landmarks like Odisha's Sun Temple and the monasteries of Sikkim.
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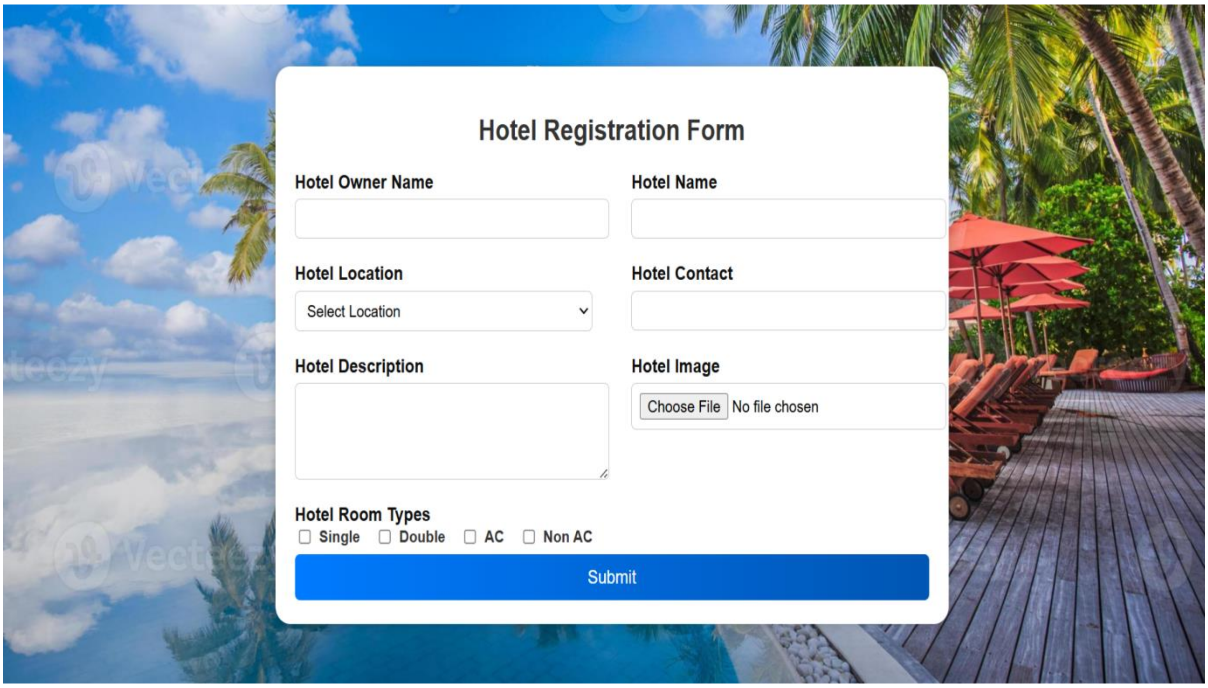
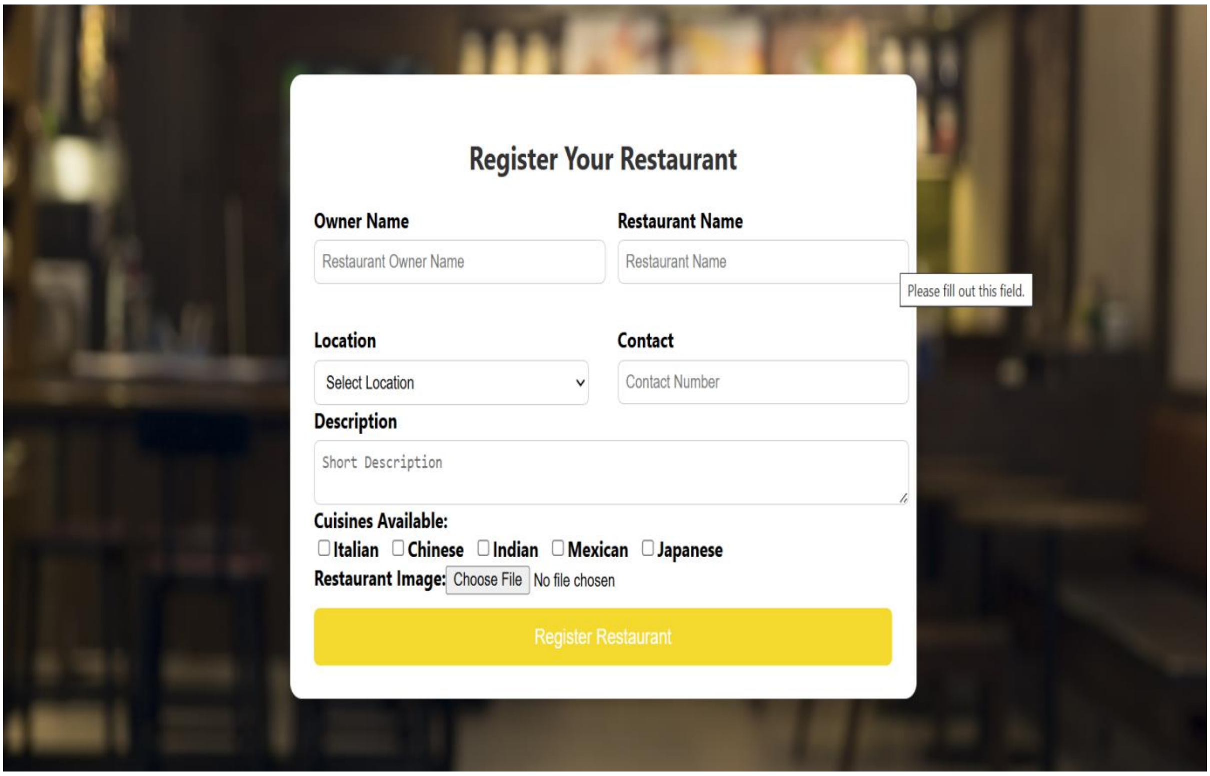
REGISTRATION FORMS






Guide Registration
Guide Name:
Location:
Select Location
Description:
Guide Image:
Choose File No file chosen
Register Guide


Darjeeling
Famous for its tea plantations, stunning Himalayan views, and toy train rides.